

What an $\text{SO}(3)$ Orientation Flow Can and Cannot Learn in Molecular-Crystal Structure Prediction

Anonymous authors

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Abstract

Rigid-body factorization is an appealing route to molecular-crystal structure prediction: freezing each molecule’s conformer reduces a crystal to a lattice, a set of fractional centroids on the torus $\mathbb{T}^3$, and a per-molecule orientation on $\text{SO}(3)$. Whether the orientation field is actually learnable on real molecular crystals has not been examined. We introduce SymMC-Flow, a space-group-conditioned rigid-body flow on $\text{lattice} \times \mathbb{T}^3 \times \text{SO}(3)$, used as a diagnostic study on 1,127 crystals from the Cambridge Structural Database. The lattice and centroid flows learn normally, but the absolute per-molecule orientation flow remains at its predict-zero floor. We trace this to a decomposition of the orientation target, $R_m = \text{rot}(g_m) \cdot R_{\text{asym}}$, into a space-group-determined relative rotation between symmetry copies and the asymmetric unit’s gauge-arbitrary free orientation, which is not learnable from composition and packing; the relative part is. Re-gauging to cancel the free part lifts held-out non-reference orientation loss by 27% (versus $\sim 0\%$ for the absolute target) and reconstructs 13.7% (mean over three seeds) of held-out multi-copy packings exactly under StructureMatcher, against 0% for the predict-floor baseline. Conditioning on a discrete space-group coset identity raises the learnable signal to roughly 48% (three-seed mean), showing the residual ceiling is set by the difficulty of *inferring* the symmetry operation from packing rather than by network capacity. End-to-end joint generation does not yet reconstruct packings exactly, placing this as a characterization rather than a finished generator. The result maps what a rigid-body $\text{SO}(3)$ flow can and cannot contribute to molecular-crystal generation, and suggests such generators should condition on crystal symmetry for the relative orientation and sample, rather than regress, the remaining free pose.

1 Introduction

Crystal structure prediction (CSP) asks for the stable periodic arrangement of a given chemical system, a problem whose combinatorial landscape has historically demanded large ensembles of density-functional relaxations. Generative models have emerged as a fast alternative that learns the distribution of stable structures directly. The progression from the crystal diffusion variational autoencoder (Xie et al., 2022) to joint equivariant diffusion over lattices and fractional coordinates (Jiao et al., 2023) and to Riemannian flow matching for materials (Miller et al., 2024) has steadily lowered both sample cost and error on inorganic benchmarks, where each atom is an independent site.

Sampling cost is a practical constraint for CSP, where a single composition may require thousands of candidate packings before relaxation and ranking. Diffusion samplers typically integrate hundreds to a thousand steps per structure, whereas flow matching learns a straight conditional path that off-the-shelf ordinary differential equation solvers traverse in tens of steps (Lipman et al., 2023; Chen & Lipman, 2024). The same advantage drove the move from $\text{SE}(3)$ diffusion to $\text{SE}(3)$ flow matching for protein backbones, where roughly five-fold fewer steps were reported at equal or better quality (Yim et al., 2023b;a). A factorized, flow-matched generator that retains this efficiency while respecting crystal symmetry is therefore an attractive target, provided each of its degrees of freedom is in fact learnable.

Molecular crystals are harder. A molecular crystal couples flexible intramolecular degrees of freedom to the rigid-body packing of whole molecules, and the most common organic space groups place several symmetry-related copies of a molecule in each cell, a regime where packing search has long been a bottleneck in community blind tests (Hunnisett et al., 2024) and a target for data-efficient learned models (Wengert et al., 2021). Two design choices have emerged. All-atom generators predict every atomic coordinate jointly: OXtal (Jin et al., 2025) learns the conditional distribution over conformation and packing with a lattice-free diffusion model, and PackFlow (Subramanian et al., 2026) flows heavy-atom Cartesian coordinates and lattice parameters with a reinforcement learning alignment step. Rigid-body generators instead treat building blocks as rigid and predict only their pose. MOFFlow (Kim et al., 2025) demonstrates this for metal-organic frameworks, casting metal nodes and organic linkers as rigid bodies and applying SE(3) flow matching to their roto-translations, which mirrors generative modeling on the rotation group (Leach et al., 2022) and the frame-based generation developed for protein backbones (Yim et al., 2023b;a).

Rigid-body factorization is attractive for molecular crystals: it collapses dozens of atomic coordinates per molecule to a single pose, exploits the modularity of the cell, and supports the short sampling trajectories characteristic of flow matching (Lipman et al., 2023; Chen & Lipman, 2024). It also introduces a per-molecule orientation variable on the rotation group SO(3). Whether that orientation field is learnable on real molecular crystals is an open question. MOFFlow reports a working method but does not analyze the orientation field in isolation, and the all-atom approaches sidestep orientation altogether. No prior work asks what a rigid-body SO(3) orientation flow actually learns on molecular crystals, or where it fails.

This question has a classical antecedent. Established molecular-CSP codes do not search the orientation of every symmetry copy independently; they search only the contents of the asymmetric unit and then *generate* the remaining copies by applying the space-group operators, so the relative pose between copies is fixed by symmetry and never a free variable (Hunnisett et al., 2024; Wengert et al., 2021). The decomposition we develop below is, in effect, a rediscovery of that division of labor for a learned flow: the symmetry-relative rotation that classical search obtains for free is what the flow can learn, and the asymmetric unit’s free pose, which classical search does sample, is what the flow cannot regress. Framing the result this way both credits the long-standing classical practice and clarifies why a naive per-copy orientation flow is solving a problem that symmetry already solves in part.

This paper answers that question. We build SymMC-Flow, a space-group-conditioned rigid-body flow on $\text{lattice} \times \mathbb{T}^3 \times \text{SO}(3)$, and train it on a corpus of 1,127 molecular crystals exported from the Cambridge Structural Database (CSD) (Groom et al., 2016). The lattice and centroid flows learn; the absolute per-molecule orientation flow does not, sitting at the predict-zero floor across two corpus sizes and after a gauge correction. We explain this with a decomposition of the orientation target and a chain of controlled diagnostics, and we show that the learnable component is real geometry, not a loss artifact.

Our contributions are:

- A rigid-body SO(3) flow for molecular crystals with a molecule-intrinsic orientation gauge that removes the spurious cross-crystal reference frame.
- The decomposition $R_m = \text{rot}(g_m) \cdot R_{\text{asym}}$ and a falsification chain (predict-zero floor, gauge fix, scaling, clean-packing control, relative re-gauge) that isolates the learnable symmetry-relative orientation from the gauge-free asymmetric-unit pose.
- An orientation-isolated StructureMatcher metric showing the learned relative orientation reconstructs 13.7% (mean over three seeds) of held-out multi-copy packings exactly, against 0% for the floor.
- Evidence from discrete coset-identity conditioning that the residual ceiling is inference-limited, not representational.

Table 1: Held-out per-head flow-matching loss, untrained $\rightarrow$ trained, on the multi-copy CSD corpus under noised conditioning and a common split. Lattice and centroid losses are independent of the orientation target; the orientation row is the absolute target R_m . Lattice and centroid learn; the absolute orientation target stays at its predict-zero floor.

head	untrained	trained	change
lattice	1.24	0.08	+94%
centroid	0.36	0.25	+30%
orientation (R_m)	5.37	5.18	+3.3%

2 Results

2.1 Rigid-body factorization and the flow

SymMC-Flow represents a molecular crystal as a lattice $L \in \mathbb{R}^{3 \times 3}$, and, for each molecule m , a fractional centroid $c_m \in \mathbb{T}^3$ and an orientation $R_m \in \text{SO}(3)$ acting on a shared, centered body-frame conformer local_m . Atom positions follow

$$\text{cart}_{m,a} = c_m L + R_m \text{local}_{m,a}, \quad (1)$$

so every copy of a chemical species shares one conformer and differs only by its pose. A conditional flow-matching objective (Lipman et al., 2023) is trained jointly on the three manifolds: a Euclidean path for a volume-normalized lattice parametrization, a torus geodesic for centroids, and an $\text{SO}(3)$ geodesic for orientations, with the vector field conditioned on the space group. An $E(n)$ -equivariant encoder (Satorras et al., 2021) embeds the conformer and pair-bias attention mixes molecules within the cell. Architecture, manifold operations, and data are detailed in Methods.

Each manifold is given a parametrization that makes its conditional path well-conditioned. The lattice is not flowed as a raw 3×3 matrix, which mixes scale and shape and admits negative determinants, but in a ten-dimensional space of a per-atom log-volume and a determinant-one shape, with the prior volume matched to the corpus mean so that the generated cells start at a physical density. Centroids live on the periodic torus and are flowed along its geodesics with wrap-around, so the target velocity respects the minimum-image convention. Orientations are flowed along $\text{SO}(3)$ geodesics, with the network predicting a body-frame angular velocity and the sampler advancing by the exponential map, which keeps every intermediate state on the manifold. This separation matters for the analysis: the three heads can be read independently, so a failure localized to orientation is visible against a backdrop of lattice and centroid that train normally.

2.2 Lattice and centroid flows learn on real molecular crystals

The corpus is a seeded random sample of the CSD, scanning 14,874 entries and retaining 1,127 ordered organic molecular crystals that factorize into 5,048 rigid blocks after the rigid-conformer and disorder gates (Methods). We establish first that the pipeline learns the packing degrees of freedom. Table 1 reports the held-out per-head flow-matching loss before and after training. The lattice loss falls by 94% and the centroid loss by 30%, both converging within the first epoch. The absolute orientation loss moves by only 3.3% and shows no downward trend over training. Lattice and centroid behave as they do on inorganic benchmarks, where every block is a single atom; orientation is the outlier, and it is the one degree of freedom that the single-atom benchmarks never exercise.

2.3 The absolute orientation flow sits at a predict-zero floor

The orientation loss equals the expected squared tangent velocity $\mathbb{E}\|u_R\|^2 \approx 5.24$ that a network outputting zero would incur. A trained orientation head that never beats this value has learned nothing. Three controls rule out the obvious explanations. First, the floor is unchanged when the corpus is scaled from 250 to 1,127 crystals, so it is not data sparsity. Second, a molecule-intrinsic orientation gauge (Methods), which removes the arbitrary cross-crystal reference frame that would otherwise make R_m a rotation relative to

some molecule in an unrelated structure, is necessary for a well-posed target but does not lift the floor. Third, a clean-packing control feeds the field the true (un-noised) lattice and centroids while still noising the orientation and keeping the orientation target, which is the best case for any two-stage scheme that generates packing first and orientation second. Orientation still moves by only 3.3% (Table 2, top row), so noised conditioning is not the cause and a two-stage architecture cannot help. The obstruction is in the target itself.

The gauge correction deserves emphasis because it is a prerequisite for the target to be well-posed at all. Without it, a species’ body frame is set by whichever copy is parsed first anywhere in the dataset, so R_m measures rotation relative to a molecule in an unrelated crystal, a quantity that is random across the corpus and would floor any orientation head by construction. Fixing the frame to a molecule-intrinsic quantity (Methods) removes this artificial randomness, and the harder, well-posed targets it produces require a lower learning rate to avoid an early-training divergence in the shared trunk. With both in place the orientation head is given every chance to learn, yet the trained loss continues to oscillate within a narrow band around the predict-zero value with no downward trend, while lattice and centroid converge in the same run. The failure is specific and reproducible, not an optimization accident.

2.4 The orientation target decomposes into a learnable and a free part

Consider a molecule m generated from an asymmetric-unit molecule by a space-group operation with rotation part $\text{rot}(g_m)$. Its absolute pose factors as

$$R_m = \text{rot}(g_m) \cdot R_{\text{asym}}, \quad (2)$$

where R_{asym} is the orientation of the asymmetric unit in the cell. The first factor is determined by the space group and is shared structure across crystals; the second is a free, per-crystal choice that dominates the target. Because the body-frame gauge is arbitrary, R_{asym} carries no consistent training signal tied to composition or packing, which produces the floor. The empirical distribution bears this out: the absolute reference-copy orientations track the Haar-uniform density on $\text{SO}(3)$ across the full range of geodesic angles, with mean 122.6° against the Haar expectation of 126.5° (Appendix B, Figure A1). A Kolmogorov–Smirnov test detects only a slight deviation from uniformity ($D = 0.04$), confined to a small near-identity excess attributable to the gauge-defining copies rather than to a broad preferred orientation. We therefore state the claim narrowly: R_{asym} is not predictable from the conditioning available here, not that orientation is unlearnable in principle. The distinction matters. A uniform *marginal* does not by itself establish that R_{asym} is conditionally independent of the packing; a weak dependence $p(R_{\text{asym}} \mid \text{packing})$ can coexist with a Haar marginal. Our claim is the operational one that no exploitable signal survives our composition-and-packing conditioning. We test this directly rather than infer it from the marginal alone: a supervised probe trained to predict R_{asym} from the lattice, the centroid arrangement, the conformer shape, and the space group does not improve on the best feature-free constant and sits at the no-information level (Appendix C), so a uniform marginal here reflects genuine conditional unpredictability rather than masking a hidden dependence. We distinguish this from an under-featurized model: a richer descriptor of the local environment, such as explicit hydrogen-bond topology or dipole alignment, could in principle expose residual signal that the present conditioning does not.

To isolate the learnable factor we re-gauge each crystal so that the first copy of every species is a reference with $R = I$ and the remaining copies carry only the relative rotation $R'_m = R_m R_0^{-1}$, which cancels R_{asym} (Methods). The transformation leaves the reconstructed crystal unchanged. To avoid crediting a trivial “predict identity” solution, we split the orientation loss into reference copies (target I) and non-reference copies (the symmetry-determined targets), and report the latter. Table 2 gives the full 2×2 over target and conditioning. The absolute target floors in both conditioning regimes. The relative target descends by 27% on held-out non-reference copies and generalizes, with an identical result under noised and clean conditioning. Across three random seeds the non-reference drop is $27.5 \pm 2.7\%$ (mean $\pm$ s.d.), so the effect is stable. Because the conformer registry is shared across crystals, a random split could in principle let a species seen in training reappear in validation; we therefore repeated the three-seed measurement under a stricter species-grouped split that assigns every crystal sharing a species to the same side of the partition (Methods). The non-reference drop is then $28.5 \pm 2.9\%$, statistically indistinguishable from the random-split

Table 2: Held-out non-reference orientation loss, untrained $\rightarrow$ trained, across orientation target and conditioning. The absolute pose floors; the relative (symmetry-determined) orientation is learnable and robust to conditioning noise.

target	noised conditioning	clean packing
absolute R_m	5.37 $\rightarrow$ 5.18 (+3.3%)	5.35 $\rightarrow$ 5.18 (+3.3%)
relative R'_m	5.37 $\rightarrow$ 3.91 (+27.1%)	5.37 $\rightarrow$ 3.94 (+26.7%)

value, so the learnable relative signal is not an artifact of species leakage between training and validation (Appendix E). The signal therefore rides on the space group, which is conditioned directly, together with the coarse centroid arrangement, and is robust to conditioning noise.

The relative analysis runs on the 1,095 crystals that contain at least two copies of a species, the subset in which a non-trivial relative target exists; their species multiplicities are dominated by two- and four-copy cells, with a tail to sixteen. On this subset the overall orientation loss, counting both copy classes, improves by 34 to 36%, and the reference copies, whose target is the identity, improve by 61%. The non-reference figure of 27% is the conservative one, since it cannot be earned by collapsing to the identity, and it is the number we carry forward. Read together, the floor of the absolute target and the descent of the relative target confirm Eq. (2) quantitatively: the only learnable content in R_m is the part that survives cancellation of R_{asym} .

2.5 Relative orientation reconstructs real packings

A drop in flow-matching loss need not correspond to correct geometry. We therefore evaluate the learned orientation by reconstruction. Holding the true lattice, centroid, and conformer fixed, we sample only the $\text{SO}(3)$ field and rebuild each crystal, then test whether it matches the ground truth under pymatgen’s StructureMatcher (Ong et al., 2013). The symmetry-induced relative rotations are large, clustering near 180° because the dominant organic space groups, such as $P2_1/c$ and $P\bar{1}$, relate copies by two-fold rotations and inversions (Groom et al., 2016), so a single deterministic integration of the field collapses to the conditional mean and underperforms. We therefore draw eight orientation samples per crystal and take the best, the appropriate generative read for a multimodal target. Reporting a controlled best-of- k match alongside the geodesic orientation error, rather than a single headline rate, follows recent cautions on match-rate metrics (Martirosyan et al., 2025).

Table 3 reports the outcome on 131 held-out multi-copy crystals, averaged over three seeds. The oracle, which uses the true orientation, matches every structure, confirming the metric’s tolerances. The trained flow reconstructs $13.7 \pm 2.0\%$ (mean $\pm$ s.d.; 12.2, 13.0, 16.0% across seeds) of held-out packings exactly, against 0% for a random (predict-floor) field and 0.8% for the naive choice of leaving every copy un-rotated. Its mean non-reference orientation error is 68° , below the 89° of a random field (Figure 1a). The 27% loss reduction is real reconstructed geometry. The reconstruction is not an artifact of the chosen tolerances: sweeping the StructureMatcher length, site, and angle thresholds moves the rate monotonically from 5.3% at the tightest setting to 20.6% at the loosest, with the oracle matching every structure throughout, and a single deterministic draw (match@1) still reconstructs a nonzero fraction against the 0% floor (Appendix D). Under the species-grouped split above, an orientation-isolated read reconstructs 10.7% of the grouped held-out crystals, confirming the reconstruction survives the leakage control.

2.6 The ceiling is inference-limited, not representational

Two experiments locate the remaining gap. Increasing model capacity and training, a wider network with more attention and encoder layers trained for more than six times as many steps, raises the non-reference signal from 27% to 45%. This is real headroom, but it stops short of the 50.6% reached by simply telling the base model which symmetry coset each copy belongs to (below), an oracle conditioning obtained without any extra capacity. Scaling the network therefore narrows but does not close the gap, and is matched by a much smaller model handed the discrete symmetry label; the relative target is not merely under-fit.

Table 3: Orientation-isolated reconstruction on 131 held-out multi-copy crystals (true lattice, centroid, and conformer; only SO(3) sampled; best of eight draws; mean over three seeds). The learned field reconstructs packings the floor and naive baselines cannot.

condition	match rate	non-ref. orient. error
oracle (true orientation)	100.0%	0°
identity (naive, $R = I$)	0.8%	163°
untrained (predict-floor)	0.0%	89°
trained	$13.7 \pm 2.0\%$	68°

Table 4: Held-out non-reference orientation loss under model and conditioning changes. Coset conditioning nearly doubles the learnable signal; the paired control isolates the effect.

configuration	non-ref. loss	change
baseline (relative target)	$5.37 \rightarrow 3.91$	+27.1%
+ capacity and training	$5.42 \rightarrow 2.98$	+45.0%
+ coset conditioning	$5.36 \rightarrow 2.65$	+50.6%
coset control (no coset)	$5.37 \rightarrow 3.91$	+27.1%

The decisive test conditions the field on a discrete space-group coset identity. For a copy generated by a crystallographic operation, the relative rotation R'_m equals that operation’s rotation part, so within a space group the non-reference relative rotations fall into a small discrete set of cosets. Clustering the observed relative rotations by geodesic distance within each space group yields a codebook of 707 cosets across the corpus, and we condition each molecule on its coset index, a discrete label of the kind used in symmetry-conditioned crystal generation (Jiao et al., 2024; Kelvinius et al., 2025; Levy et al., 2025), rather than the continuous rotation itself. Table 4 shows that coset conditioning raises the non-reference signal to $47.9 \pm 3.2\%$ over three seeds (50.6% on the single run shown in the table), with reference copies reaching 89%, nearly doubling the learnable component relative to a paired control that reproduces the 27% baseline on the identical corpus and split (Figure 1b). Most of the ceiling is therefore the difficulty of inferring which symmetry operation generated each copy from the noised packing, not an inability of the SO(3) head to represent the rotation. The residual that remains even with the coset supplied is attributable to the free R_{asym} and to the axial gauge freedom of symmetric-top molecules, whose orientation about a symmetry axis is genuinely unobservable.

This experiment is a representability diagnostic, not a deployable conditioning. Because the codebook is built by clustering the observed relative rotations, the coset index is derived in part from the target it helps predict, so the 50.6% is an upper bound on what the discrete symmetry identity can contribute, isolating representability from inference. The same label is in principle available at sampling time without the answer, since the coset is a deterministic function of the space group, the Wyckoff assignment, and the copy ordering, which template-based generators already provide (Jiao et al., 2024; Kelvinius et al., 2025; Levy et al., 2025); realizing that gain in a sampler is left to future work.

2.7 End-to-end generation is bottlenecked by the joint packing

The orientation-isolated metric supplies the true lattice and centroid. The complementary question is how the full pipeline performs when all three manifolds are sampled from the prior, that is, end-to-end rigid-body structure prediction given only the molecular composition and conformer. Generating complete crystals for the held-out molecules (match@20, three seeds) yields no exact match at the StructureMatcher tolerances above (0%, Wilson 95% CI 0–2.8%, $n = 131$); the single-draw rate (match@1) is likewise 0%, so the gap is not an artifact of the best-of- k read. The best-of-20 per-component errors locate the bottleneck: a median lattice-parameter error of 0.46, a fractional centroid error of 0.35, and an orientation error of 57° (medians over best-of-20 draws; this de-novo orientation error is not directly comparable to the orientation-isolated 68° of Table 3, which is a mean over best-of-8 draws taken with the true packing supplied). Every head

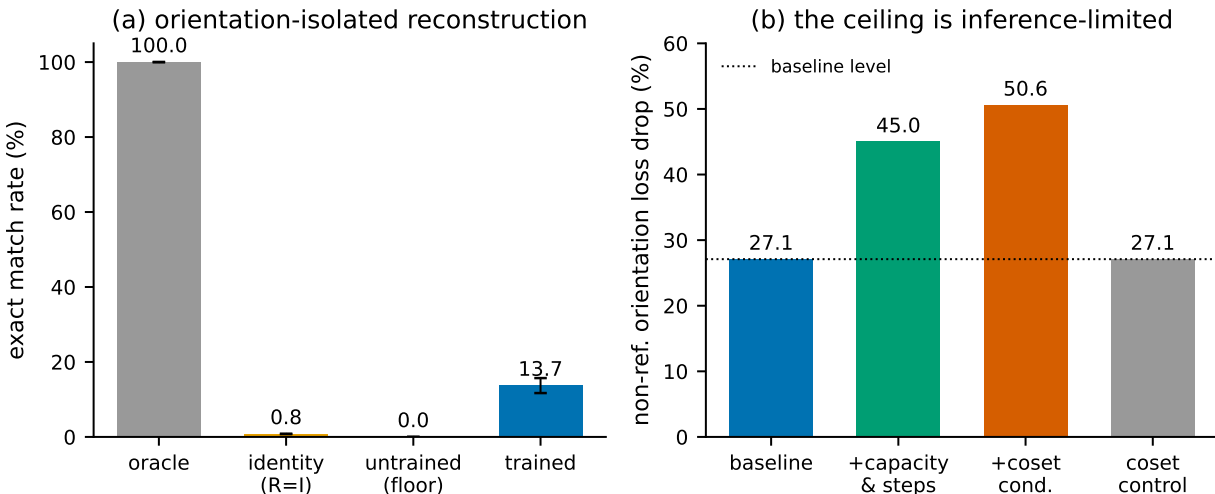


Figure 1: The orientation results. (a) Orientation-isolated StructureMatcher reconstruction rate by condition (true lattice, centroid, and conformer; only $SO(3)$ sampled; best of eight draws; $n = 131$ held-out crystals; mean over three seeds). The trained flow reconstructs 13.7% of packings exactly, against 0% for the predict-floor field and 0.8% for the naive $R = I$ baseline. (b) Held-out non-reference orientation loss drop under the strengthening levers; coset conditioning nearly doubles the learnable signal over the baseline level (dotted), while the no-coset control reproduces it, showing the ceiling is inference-limited rather than representational.

contributes; in particular the centroid field, which improves by only 30% in loss, leaves molecules displaced by roughly a third of a cell, and the orientation still carries the gauge-free component. Scaling does not lift the joint result: the higher-capacity model above, despite sharper components (median lattice-parameter error 0.41, centroid 0.35, orientation 50°), still yields 0% end-to-end at both match@1 and match@20. To test whether this bottleneck is specific to the rigid-body factorization or instead reflects the corpus scale, we trained an all-atom control on the same corpus and split in which every atom is its own block and the orientation head is disabled, the representation used by all-atom generators (Jin et al., 2025; Subramanian et al., 2026). Despite its lattice and centroid heads training normally (an 88% drop in flow-matching loss), it likewise reconstructs no held-out crystal exactly (match@1 and match@20 both 0%, $n = 131$), equal to its own random-prior floor. Its best-of-20 packing-component errors are as large as the rigid-body model’s (median lattice-parameter 0.33 and fractional-coordinate 0.46; Appendix F), locating the failure in the same joint packing. End-to-end reconstruction at this corpus scale is therefore out of reach for the all-atom representation as well, so the 0% is a property of the scale and compute budget rather than a deficiency unique to rigid-body factorization. This control is our own all-atom flow trained on the same thousand-crystal corpus, not one of the published all-atom generators (Jin et al., 2025; Subramanian et al., 2026), which are trained on far larger datasets with substantially more compute; the comparison therefore shows that the all-atom *representation at this scale* does not rescue end-to-end generation, not that those larger models would fail. The orientation-isolated 13.7% is therefore an upper bound that is attainable only when the packing is supplied, and end-to-end molecular-crystal generation at this corpus scale and compute budget remains open. This is consistent with the scope of the paper: the contribution is the characterization of the orientation field, not a finished generator.

2.8 Method validation on inorganic benchmarks

For completeness we validated the SymMC-Flow architecture on standard single-block inorganic benchmarks, where orientation is inactive. On carbon-24 and MP-20 the method is competitive with joint equivariant diffusion (Jiao et al., 2023) at roughly an order of magnitude fewer sampling steps, winning some match cells and losing others. Full numbers and the head-to-head protocol are given in Appendix A; these results confirm the flow and sampler are sound and are not the focus of this study.

3 Discussion

The orientation field of a rigid-body molecular-crystal flow is neither fully learnable nor uniformly broken. It splits into a space-group-determined relative orientation that a flow learns and reconstructs, and a gauge-free asymmetric-unit orientation that the model cannot predict from composition and packing. This split is a property of the representation, not of the optimizer or the architecture, which is why it survives gauge correction, data scaling, capacity increases, and clean conditioning.

The decomposition reconciles the divergent design choices in the field. All-atom generators (Jin et al., 2025; Subramanian et al., 2026) never expose R_{asym} as an explicit target and so do not encounter the floor, trading the low dimensionality of the rigid-body representation for a target without a gauge obstruction. MOFFlow (Kim et al., 2025) succeeds with a rigid-body SE(3) flow in part because framework connectivity constrains block orientations far more tightly than the free pose of a molecule in a van der Waals packing, shrinking the gauge-free component. The present analysis predicts that rigid-body SO(3) generation will be most effective precisely where orientation is constrained by the surrounding network or by high site symmetry, and least effective for low-symmetry organic packings.

On evaluation, the gap between the flat orientation loss and the 13.7% reconstruction rate shows why a single match-rate number is an unreliable summary (Martirosyan et al., 2025): the learnable signal is visible only when the metric isolates orientation and samples the multimodal target. We report the geodesic error and a controlled best-of- k match for this reason.

The coset result also carries a practical implication for how rigid-body generators should be conditioned. Template-based pipelines already supply a space group and Wyckoff assignment at sampling time, and the coset index is a deterministic function of that assignment together with the copy ordering. A generator that consumes this discrete label, as space-group- and Wyckoff-conditioned models already do for atomic sites (Jiao et al., 2024; Kelvinius et al., 2025; Levy et al., 2025), converts the inference-limited part of the orientation field into a near-deterministic lookup, leaving only the free pose to be sampled. This suggests that the right division of labor for rigid-body molecular CSP is to condition on symmetry for the relative orientation and to sample, rather than regress, the remaining gauge-free degree of freedom.

This study has limits. End-to-end generation does not yet reconstruct held-out packings exactly, and the analysis isolates orientation rather than delivering a finished generator. A de-novo stable-unique-novel evaluation would not add much here either, because the universal interatomic potentials used to judge stability are trained on inorganic data (Deng et al., 2023) and do not transfer to organic molecular crystals; we therefore evaluate by reconstruction against known structures. The corpus is roughly one thousand crystals and training is at modest scale, so the absolute generation numbers should be read as a lower bound that more data and compute would improve. The rigid-conformer gate excludes flexible molecules by construction, and the conformer registry is shared across crystals, so a species whose conformer differs between polymorphs beyond tolerance is rejected. The corpus is therefore one of rigid, largely organic (C/H/N/O-dominated) molecules, and the conclusions should be read with that scope. The unlearnability statement is likewise scoped to molecules on *general* positions, where the asymmetric-unit pose is genuinely free; a molecule sitting on a special position carries a site symmetry that constrains its orientation, and that constraint is in principle learnable, consistent with the prediction above that rigid-body SO(3) generation is most useful where site symmetry is high. These choices keep the rigid-body benchmark honest but narrow its coverage.

Two directions follow. A hybrid that predicts a rigid pose and then refines the conformer would extend the approach to flexible molecules while keeping the fast pose flow. Conditioning the orientation field on Wyckoff or coset information at sampling time, suggested by the coset experiment, would convert the inference-limited component into a usable signal whenever the symmetry assignment is available, as it is in template-based generation.

4 Methods

4.1 Rigid-body factorization and intrinsic gauge

Each crystal is parsed into rigid molecular blocks by a covalent-bond graph with periodic-boundary unwrapping. Molecules of one species are keyed by a Weisfeiler–Lehman hash of the element-labelled bond graph and share a single reference conformer; atom correspondence between a copy and its reference is the minimum-RMSD graph isomorphism, so a molecule’s own automorphisms inject no spurious rotation. The reference conformer is rotated into a molecule-intrinsic frame given by the principal axes of its gyration tensor, with per-axis sign fixed by an element-weighted third moment and the frame forced right-handed. This removes the arbitrary cross-crystal reference that otherwise makes R_m a rotation relative to an unrelated molecule, the gauge correction referenced in Results. The pose R_m of each copy is then the Kabsch rotation aligning it to the intrinsic frame. Lattices are rebuilt from their six cell parameters to a canonical frame.

4.2 Manifolds and flow matching

The lattice is flowed in a ten-dimensional parametrization of per-atom log-volume and a determinant-one shape, with a straight-line conditional path. Centroids follow torus geodesics; orientations follow $\text{SO}(3)$ geodesics with the tangent velocity as target. We use conditional flow matching (Lipman et al., 2023) with optimal-transport coupling of the exchangeable prior and data within each crystal (Pooladian et al., 2023). Sampling integrates the field from the prior to the data manifold with a deterministic Runge–Kutta scheme, using the $\text{SO}(3)$ exponential map for the orientation update; molecular-crystal samples used here take fifty steps, approximately 0.26 s per structure on a single RTX 5090 GPU in batches of 32, consistent with the order-of-magnitude step-count reduction that motivates flow matching over diffusion. The construction follows Riemannian flow matching (Chen & Lipman, 2024) and parallels frame-based generative models for proteins (Yim et al., 2023b;a) and metal–organic frameworks (Kim et al., 2025). The orientation target is the constant-velocity geodesic tangent $u_R = \log(R_0^\top R_1)$ carried in the body frame, where R_0 is the prior and R_1 the data orientation, and the loss is the squared geodesic residual between the predicted and target tangents averaged over real molecules.

4.3 Predict-zero floor and clean-packing control

The predict-zero floor is the loss of the trivial field that outputs the zero velocity, $\mathbb{E}\|u_R\|^2$, estimated by averaging the squared target tangent over the validation set; it is the value a head must beat to have learned anything, and on this corpus it is approximately 5.24. The clean-packing control replaces the noised conditioning state with the true lattice and centroids while leaving the orientation input noised and the orientation target unchanged, so no label information leaks. Under this control the lattice and centroid heads see a state inconsistent with their own velocity targets and their losses are not interpretable, which is why only the orientation number is read from that run.

4.4 Network

An $E(n)$ -equivariant graph network (Satorras et al., 2021) encodes each conformer into an invariant token. Tokens are augmented with the current centroid, orientation, lattice, time, and a space-group embedding, then mixed by pair-bias attention with pairwise features built from torus and Cartesian displacements. Three heads predict the lattice, centroid, and orientation velocities. The centroid field can be symmetrized over the space-group action. For the coset experiment a per-molecule coset embedding is added to each token when enabled.

4.5 Relative gauge and coset codebook

The relative gauge sets the first copy of each species to $R = I$ and replaces the others by $R'_m = R_m R_0^{-1}$, with the shared conformer transformed by R_0 so that reconstruction is unchanged; a reference mask records which copies are trivial. The coset codebook clusters non-reference relative rotations within each space group by geodesic distance into a shared set of discrete indices, which become the per-molecule coset labels.

4.6 Training

Models are trained with Adam at a learning rate of 3×10^{-4} , gradient clipping, and a batch size of 16. The reduced learning rate, relative to a common default of 10^{-3} , is needed because the intrinsic-frame orientation targets produce larger early gradients that otherwise diverge in the shared trunk within the first tens of steps. The lattice prior volume per atom is matched to the corpus mean so that generated cells begin at a physical density. Held-out losses are averaged over several independent draws of the flow time and prior at a fixed seed, so that the untrained and trained measurements are taken under identical random conditions and their difference reflects learning rather than sampling noise. The baseline relative run uses the default network ($d=128$, four attention layers, four encoder layers) for 800 steps; the capacity run widens the model to $d=256$ with six attention and five encoder layers and trains for 5,000 steps. A non-finite-step guard skips any optimizer update whose loss or gradient norm is not finite, so a transient ill-conditioned batch in the higher-capacity model cannot corrupt the weights; the run reported here skipped a single step.

4.7 Species-grouped split and all-atom baseline

The species-grouped split guards against leakage from the shared conformer registry. We build an undirected graph whose nodes are crystals and whose edges join any two crystals that share a chemical species, take its connected components by union-find, and assign whole components to training or validation, so no species appears on both sides. The relative-orientation diagnostic is then rerun over three split seeds at the default network and batch size and compared to the random-split result. The all-atom baseline replaces the rigid-body factorization with one block per atom and disables the orientation head ($\lambda_{\text{orient}} = 0$), matching the all-atom representation; it is trained on the same corpus and split and evaluated for de-novo reconstruction against both its random-prior floor and the trained field.

4.8 Orientation-isolated match metric

Reconstruction holds the true lattice, centroid, and conformer fixed and integrates only the orientation field from a random $\text{SO}(3)$ prior under clean-packing conditioning. Reconstructed crystals are built with Eq. (1) and compared to the ground truth with StructureMatcher (Ong et al., 2013) using a length tolerance of 0.3, a site tolerance of 0.5, and an angle tolerance of 10° , slightly looser than the pymatgen defaults to accommodate the rebuilt cells. The oracle row, which reconstructs every structure at these tolerances, confirms they are not so loose as to admit spurious matches. For each crystal we draw eight orientation samples and count a match if any draw matches; we also report the minimum mean geodesic error over draws on non-reference copies. Reported match rates carry a binomial (Wilson) 95% confidence interval of approximately ± 6 percentage points at $n = 131$.

4.9 Data

The corpus is a seeded random sample of the CSD (database version 601, accessed via the CSD Python API) (Groom et al., 2016) filtered to ordered, organic, three-dimensional, non-polymeric molecular crystals within size and Z' bounds, yielding 1,127 crystals and 5,048 rigid blocks from 14,874 scanned entries; 1,095 contain at least two copies of a species and define the multi-copy subset used for the relative and coset analyses. The species multiplicities of that subset are concentrated at two and four copies per cell, with smaller populations at three, six, eight, twelve, and sixteen, reflecting the general-position multiplicities of the common organic space groups. A fixed seed gives a 964/131 train/validation split, used identically across every diagnostic so that comparisons are paired. CSD structures are not redistributable; a refcode manifest and the export and analysis code reproduce the dataset from a licensed CSD installation (see the Reproducibility Statement).

Reproducibility Statement

All architectural, manifold, training, and evaluation details needed to reproduce the diagnostics are specified in Methods and Appendices A–F: the flow-matching objective and $\text{SO}(3)$ /torus/lattice manifold operations, the intrinsic-gauge and relative-gauge constructions, the coset codebook, the network, the Adam hyper-

parameters and step budgets, the species-grouped split, and the StructureMatcher tolerances used for the orientation-isolated metric.

The molecular-crystal corpus is a seeded random sample of the Cambridge Structural Database (CSD) (Groom et al., 2016). The CSD is licence-restricted, so the CIF structures and the factorized tensors derived from them cannot be redistributed; this is the sole reason the primary data are not openly deposited. Reproduction on any licensed CSD installation is nonetheless fully determined: a manifest of the 3,500 CSD refcodes used to build the corpus, together with per-entry selection metadata (formula, Z' , R-factor, atom count, and space group), and the export and factorization scripts, deterministically regenerate the exact corpus and derived tensors, and a fixed random seed reproduces the identical 964/131 train/validation split used throughout. The complete SymMC-Flow implementation, the data-preparation pipeline, and this refcode manifest will be released in a public repository and permanently archived upon publication; they are withheld from this submission only to preserve double-blind anonymity, and an anonymized copy can be provided to reviewers through the OpenReview discussion on request.

The authors used an AI coding and writing assistant for software development and manuscript copy-editing; all scientific claims, experiments, and results were verified by the authors, who take full responsibility for the content.

Broader Impact Statement

This is a diagnostic study of what a generative flow can and cannot learn about molecular-crystal packing. It advances basic methodology for predicting crystal structures from molecular composition, a capability relevant to computational chemistry and materials discovery, including the screening of candidate polymorphs of organic solids. The work reports a characterization rather than a deployable generator, introduces no new released datasets, and builds no system that acts on people or that presents an evident dual-use risk; we therefore foresee no negative societal impact specific to this work beyond those broadly associated with accelerating molecular and materials design. All experiments use structures from a licensed crystallographic database under its terms of use.

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A Method validation on single-block inorganic benchmarks

The orientation analysis in the main text isolates a target obstruction that is specific to multi-copy molecular crystals. To confirm that the underlying SymMC-Flow architecture, training objective, and Runge–Kutta sampler are sound independently of that obstruction, we benchmarked the model on two standard single-block datasets, carbon-24 and MP-20, in which each atom is its own rigid block and the orientation head is inactive ($\lambda_{\text{orient}} = 0$). These results are reported for completeness; they are not the contribution of this paper.

Protocol. We compare against joint equivariant diffusion (Jiao et al., 2023) using its released models, with matched evaluation: 256 held-out targets, 20 samples per target, and the same root-mean-square structure distance routine for both methods. Match rate at k (“match@ k ”) is the fraction of targets for which at least one of k samples matches the reference under StructureMatcher (Ong et al., 2013). SymMC-Flow samples in 50 Runge–Kutta steps, against roughly 1,000 steps for the diffusion baseline.

Outcome. Table A1 shows a mixed result: each method wins two of the four match cells. SymMC-Flow leads on carbon-24 match@1 and on MP-20 match@20, while the diffusion baseline leads on MP-20 match@1 and on carbon-24 match@20 and produces tighter RMSE on matched structures. The honest summary is that SymMC-Flow is competitive at roughly an order of magnitude fewer sampling steps, not a uniform improvement. This is sufficient to establish that the flow and sampler are well-behaved, which is all the main-text analysis requires.

Table A1: SymMC-Flow versus a diffusion baseline (Jiao et al., 2023) on single-block inorganic CSP under a matched evaluation (256 targets, 20 samples, identical RMSD routine). Match rates in percent; higher is better. Each method wins two of the four cells.

dataset	metric	SymMC-Flow	diffusion baseline
carbon-24	match@1	26.7	18.8
carbon-24	match@20	79.7	89.1
MP-20	match@1	41.4	48.0
MP-20	match@20	80.5	76.2

Caveat on benchmarks. Recent work shows that carbon-24 contains a large fraction of duplicate structures and that match-rate metrics can mislead when identical building blocks exhibit structural variety (Martirosyan et al., 2025). The numbers above should be read as a sanity check on the architecture under the field’s conventional protocol rather than as a definitive ranking.

B Is the absolute molecular orientation uniform on SO(3)?

The main-text claim that the free asymmetric-unit pose R_{asym} carries no learnable signal predicts that the absolute per-molecule orientations should be close to Haar-uniform on SO(3), since a uniform target offers nothing to predict. For Haar-uniform rotations the geodesic angle θ from the identity has density $p(\theta) = (1 - \cos\theta)/\pi$ on $[0, \pi]$, with mean 126.5° . Figure A1 compares the empirical distribution of the reference-copy orientations, which equal R_{asym} up to the operation generating the first-detected copy, to this density. The empirical distribution tracks the Haar curve across the full angular range, with mean 122.6° ($n = 1,467$). A one-sample Kolmogorov–Smirnov test against the Haar law gives $D = 0.04$, a small but nonzero deviation localized to a near-identity excess that comes from the gauge-defining copies, whose self-alignment lands near the identity. The absolute orientation is therefore close to uniform, consistent with the predict-zero floor, and the residual deviation is a gauge artifact rather than a broad packing preference.

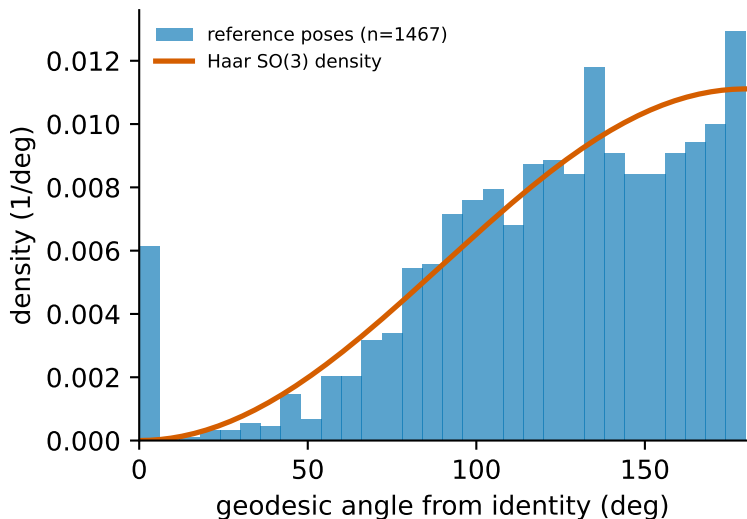


Figure A1: Empirical distribution of absolute reference-copy orientations (geodesic angle from the identity) against the Haar-uniform $SO(3)$ density. The two agree across the range apart from a small near-identity excess from gauge-defining copies.

C Is R_{asym} conditionally predictable from packing?

The uniformity test of the previous section concerns the *marginal* distribution of R_{asym} . A uniform marginal is necessary but not sufficient for the operational claim that packing carries no signal about the free pose: a weak conditional dependence $p(R_{\text{asym}} \mid \text{packing})$ can coexist with a Haar marginal once the packing is marginalized out. We therefore test the conditional directly with a supervised probe. For each species we form a feature vector from the lattice cell parameters, the molecule’s fractional centroid, the mean and spread of all centroids in the cell, the conformer’s intrinsic shape (the gyration-tensor eigenvalues), the molecule and atom counts, and a learned space-group embedding, and we train a three-layer network to regress the reference-copy orientation (a continuous six-dimensional rotation parametrization with a Frobenius loss). The probe sees only composition and packing; the orientation is never an input.

Table A2 reports the held-out geodesic error against two references: the single best feature-free predictor, which is the Fréchet (chordal) mean of the training orientations, and a random-rotation guess. Averaged over three splits the probe attains 125.2° , which does not improve on the feature-free constant at 122.4° and is statistically indistinguishable from the random-guess level of 123.5° ; the probe’s mean “improvement” over the constant is -2.4% , i.e. no improvement. The network can drive its training error well below these values, so the failure is one of generalization, not of capacity. The conditional test thus reaches the same conclusion as the marginal one by an independent route: under the composition-and-packing conditioning available here, R_{asym} is not predictable, which is the precise, operational sense in which we call it unlearnable.

Table A2: Conditional-predictability probe for R_{asym} : held-out mean geodesic error (degrees, mean over three splits; lower is better). The probe, given composition and packing, does not beat the feature-free constant and sits at the no-information level.

predictor	held-out geodesic error
random guess (no information)	123.5°
constant (Fréchet mean, feature-free)	122.4°
probe (composition + packing features)	125.2°

D StructureMatcher tolerance sweep for the orientation-isolated rate

The main-text reconstruction rate uses a length tolerance of 0.3, a site tolerance of 0.5, and an angle tolerance of 10° . To show the result is not an artifact of that choice, Table A3 sweeps the three StructureMatcher tolerances on a representative trained checkpoint, reporting both a single deterministic draw (match@1) and the best of eight draws (match@8). The match decision uses `StructureMatcher.fit` (maximum atomic displacement $\leq$ the site tolerance), the same strict criterion as the headline; the median root-mean-square displacement is computed only over structures that already pass. The rate rises monotonically from 5.3% at the tightest setting to 20.6% at the loosest, while the oracle that uses the true orientation matches every structure at all four settings, confirming the tolerances are not so loose as to admit spurious matches. The default row (0.3/0.5/ 10°) gives 16.0% match@8 on this checkpoint; the main-text $13.7 \pm 2.0\%$ is the mean over three independent training seeds at this row.

Table A3: Orientation-isolated reconstruction under a StructureMatcher tolerance sweep (single trained checkpoint; $n = 131$ held-out multi-copy crystals; true lattice, centroid, and conformer; only SO(3) sampled). The oracle matches every structure at every setting. Match decided by `StructureMatcher.fit`; RMSD is over matched structures only.

ℓ tol	stol	angle	match@1	match@8	med. RMSD (Å)	oracle match@8
0.20	0.20	8°	0.0%	5.3%	0.089	100%
0.30	0.30	10°	1.5%	9.9%	0.113	100%
0.30	0.50	10°	3.8%	16.0%	0.128	100%
0.50	0.70	15°	4.6%	20.6%	0.142	100%

E Leakage-controlled species-grouped split

Because a single conformer registry is shared across crystals, a random train/validation split could place two crystals of the same species on opposite sides, raising the concern that the learnable relative-orientation signal reflects species memorization rather than transferable geometry. We rule this out with a species-grouped split (Methods): crystals sharing any species are grouped by union-find and whole groups are assigned to one side, so no species crosses the partition. Table A4 compares the held-out non-reference orientation drop under the two split regimes over three seeds. The grouped split gives $28.5 \pm 2.9\%$, statistically indistinguishable from the random-split $27.5 \pm 2.7\%$, so the effect is not species leakage. An orientation-isolated reconstruction read on the grouped split (seed 0) recovers 10.7% of held-out crystals (match@8; 4.6% match@1; median RMSD 0.140 Å) against 0% for the predict-floor field, confirming the reconstruction also survives the leakage control.

Table A4: Held-out non-reference orientation drop under random versus species-grouped train/validation splits (three seeds each, mean $\pm$ s.d.). The grouped split eliminates shared-species overlap; the signal is unchanged.

split	non-reference drop
random (main text)	$27.5 \pm 2.7\%$
species-grouped (no leakage)	$28.5 \pm 2.9\%$

F All-atom de-novo baseline

To place the 0% end-to-end rigid-body result in context, we trained an all-atom control that discards the rigid-body factorization: every atom is its own block and the orientation head is disabled ($\lambda_{\text{orient}} = 0$), the representation used by all-atom molecular-crystal generators (Jin et al., 2025; Subramanian et al., 2026). It is trained on the same 964/131 split and evaluated for de-novo reconstruction (all manifolds sampled from

the prior) against both a random-prior floor and the trained field, with match@20 best-of-twenty draws. Its lattice and centroid heads train normally (an 88% drop in flow-matching loss over 8,000 steps), yet it reconstructs no held-out crystal exactly: match@1 and match@20 are both 0% (Wilson 95% CI 0–2.8%, $n = 131$), identical to the random-prior floor (Table A5). End-to-end molecular-crystal reconstruction at this corpus scale is therefore out of reach for the all-atom representation as well, so the 0% reported for rigid-body generation reflects the corpus scale and compute budget rather than a deficiency unique to the rigid-body factorization. This is our own all-atom flow at the thousand-crystal scale, not one of the published all-atom generators (Jin et al., 2025; Subramanian et al., 2026) trained on far larger corpora; it isolates the effect of the representation at a fixed scale and makes no claim about those larger models.

The best-of-twenty packing-component errors confirm the failure is the same joint packing that bottlenecks the rigid-body model, not a representation-specific pathology: the trained all-atom flow leaves a median lattice-parameter error of 0.33 and a median fractional-coordinate error of 0.46 (Table A5), comparable to the rigid-body de-novo errors reported in the main text (lattice 0.46, centroid 0.35). Both representations place molecules roughly a third to a half of a cell from their true positions, which is why neither reconstructs a crystal exactly.

Table A5: De-novo all-atom reconstruction on 131 held-out crystals (all manifolds sampled, best of twenty draws). The trained all-atom flow does not beat its random-prior floor at this corpus scale, despite its packing heads training normally, and its best-of-twenty packing errors match the rigid-body model’s.

field	match@1	match@20	med. lattice-param	med. frac. coord.
random prior (floor)	0.0%	0.0%	—	—
trained all-atom flow	0.0%	0.0%	0.33	0.46